Simon B. Laursen



Freelance Software Engineer & Architect

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Summary

Software engineer and architect with 10 years of experience building backend systems and platforms in regulated and enterprise environments, including five years as lead developer at Bankdata. I specialize in Java backend development and modernization: decoupling from legacy systems, moving to event-driven and eventually consistent architectures, and delivering on Kubernetes, AWS, and Azure.

I am currently consulting for DSV, building Java 25/Spring Boot microservices on AKS that take over business logic from legacy .NET integration services. At Bankdata, I was lead developer on the Java/Quarkus transaction engine behind Bankdata's first customer-data platform on AWS, processing 3.5 billion transactions for 1.4 million customers, and then led the enabling team building the CDC- and Kafka-based platform that decouples application teams from on-premises legacy systems.

Security runs through my work. As lead developer, I helped take Bankdata's policy-as-code authorization platform on Open Policy Agent from proof of concept to production. It was built to scale and is now a central platform component used by all teams at Bankdata. I have also worked as a security architect on threat modeling, security reviews, and Danish standardization work on IoT and OT security. I am especially motivated by products and teams in regulated environments where security is mandatory and designed in from the start, not patched on afterwards.

I hold a PhD in modeling and verification of distributed systems and serve as an accredited external examiner in Computer Science and in Electronics & IT engineering in Denmark.

Areas of Expertise

Java Backend Development · Legacy Modernization · Platform Engineering · Event-Driven Architecture · Kubernetes / OpenShift · API Security & Authorization · DevSecOps

Tools and Technologies

Languages & Frameworks	Java, Quarkus, Spring Boot, C#/.NET, Go, SQL, Bash, Rego
Data & Integration	Kafka, CDC, IBM MQ, SQS/SNS, REST APIs, DB2, PostgreSQL/Aurora, MongoDB
Platforms & Cloud	Kubernetes, OpenShift, AKS, AWS (ECS Fargate, S3), Terraform/OpenTofu, Argo CD, Linux
Security	OPA/Styra, OAuth2/OIDC, policy-as-code, threat modeling, SAST, SCA, OWASP SAMM / ASVS
Delivery	GitHub Actions, Azure DevOps, Jenkins, GitOps
AI-assisted development	GitHub Copilot, Claude Code, Codex, agentic workflows

Career and projects

DSV | Senior Java Developer (External Consultant) — Aug 2026 – Present
Remote · Integration Modernization

Modernizing legacy integration services to Java and Kubernetes

DSV is modernizing its integration landscape. BizTalk has already been moved to IBM App Connect Enterprise (ACE), and the remaining legacy .NET Framework services, hosted in IIS and as Windows services, are now moving to ACE for integration and to Java on Kubernetes for business logic.

My role: I work hands-on with the analysis, design, and implementation of Java 25/Spring Boot microservices on Kubernetes (AKS) that ACE calls for business logic, using Kafka, MongoDB, PostgreSQL, OAuth2/OIDC, and API management. A significant part of the role is reverse-engineering the existing services and integration flows to decide whether each should be migrated, replaced, moved into ACE, or decommissioned rather than simply rewritten. I design APIs and service boundaries that keep business logic in Java and integration and orchestration in ACE, and use AI-assisted and agentic development heavily to understand and migrate legacy business logic, document existing behavior, and accelerate implementation.

Impact: The work is turning a direct technology rewrite into a more selective modernization effort, identifying where dedicated microservices add value and where existing capabilities can be simplified, consolidated, or retired.

BANKDATA | Lead Software Developer — Aug 2025 – Aug 2026
Aarhus · Cloud Application Enablement

Enablement platform for decoupling from on-premises DB2 and mainframe

Bankdata's core banking data lives in centralized DB2 and mainframe systems on-premises at JN Data, and for 40 years the organization has been built around synchronous calls into them. To modernize at scale, application teams needed to build new services in the cloud without those direct dependencies. The Cloud Application Enablement team builds the shared platform that makes this possible.

My role: I led a three-person enabling team supporting more than five stream-aligned teams, and stayed hands-on in the code. I was the de facto architect for data movement, CDC, PostgreSQL in AWS, and the patterns and libraries application teams built on, and a permanent member of the Technical Advice Forum, which advised developers and architects across the organization on architecture decisions.

Architecture decisions: We documented the platform in ADRs that went through the Technical Advice Forum and were adopted widely. A key decision was to make Kafka a platform for data streaming and CDC, not a general integration platform: all data on Kafka must be re-creatable from DB2 by restarting CDC on the relevant tables. Many teams wanted Kafka for fully asynchronous communication, but we deliberately ruled that out to keep the platform simple at first.

Data and consistency: We built CDC pipelines off the core banking systems, Kafka-based data distribution, and PostgreSQL/Aurora read models in AWS that decouple application teams from the centralized on-premises databases. The hardest part was moving an organization used to synchronous calls toward an architecture that is eventually consistent by design. We established the patterns that make this work in practice: CQRS-style read models, idempotent consumers, outbox-based publishing, versioned event contracts, and automated reconciliation against DB2 as the source of truth. I designed and implemented a shared outbox library for Java/Quarkus: application data and outgoing events are stored atomically, and a database-backed relay publishes them to Kafka with safe concurrency via SKIP LOCKED, sequencing, and idempotent production.

To make the platform self-service, we wrote a Kubernetes controller in Go that provisions databases on demand, and provided golden-path templates, CI/CD, Infrastructure as Code, and shared monitoring for the data pipelines, so teams could adopt it without bespoke work.

Impact: The platform let application teams move to the cloud and modernize incrementally while the enabling team absorbed the operational complexity. It took the approach behind Bankdata's first AWS application and scaled it across the company.

BANKDATA | External Consultant
Aarhus · Transactions & Categorization
Aug 2023 – Jul 2025

Java/Quarkus transaction platform on AWS

Bankdata launched its first project with customer data hosted on public cloud (AWS): a transaction and categorization engine replacing an existing COTS solution and processing around 3.5 billion transactions for 1.4 million customers, with strict requirements for scalability, auditability, and integration with existing on-premises systems.

My role: As lead developer in a cross-functional team of eight, I led the application architecture and implementation of an event-driven Java/Quarkus solution, coordinating development and staying hands-on in the code throughout delivery. We first built a proof of scale and then Bankdata's first application with customer data in production on AWS, before a shared platform existed. Because implementation started before the target infrastructure was fixed, we kept application logic, runtime configuration, and infrastructure clearly separated, so the platform could run on-premises, hybrid, or in the cloud. It runs on ECS Fargate, SQS/SNS, S3, and Aurora PostgreSQL, with ingestion from on-premises systems via IBM MQ, and I built the CI/CD and Infrastructure as Code with GitHub Actions and Terraform.

Impact: The engine is in production and powers categorized transactions in the member banks' net and mobile banking. It established the patterns for secure cloud delivery and legacy integration, and its lessons fed into the Cloud Application Enablement data platform and into the shape of Bankdata's container platform.

BANKDATA | External Consultant
Silkeborg · AppSec / Authorization
Jun 2021 – Jul 2023

Central API authorization and policy-as-code platform

Bankdata needed central authorization for APIs and platform services, with consistent policy enforcement across teams and environments. We built a policy-as-code platform on Open Policy Agent (OPA) and Styra, tackling both the technical design and the organization-wide adoption challenge.

My role: As lead developer, and together with the manager of the Authorization area, I was responsible for the vendor engagement to plan and implement a scalable proof of concept, then helped mature the solution from alpha to production, planning the roadmap and features with stakeholders from security, business administration, and the development teams. I drove adoption of secure API patterns across the organization, mentored team members, and helped grow the team from one to five developers.

Architecture: Policies were enforced in the API gateway and by OPA sidecars that services called through an internal SDK, with Rego policies tested and released through their own CI/CD pipeline integrated with existing change management. Because the control plane was not open source, we designed it to be replaceable: the Kubernetes controller we built gave teams self-service and abstracted the control plane away, which later let Bankdata replace it three years after I left, when the vendor had been acquired and the product discontinued.

Impact: Adoption grew from 2–5 teams at the start to 20–30 teams when I left, and today all of Bankdata's roughly 1,000 developers use the platform. It became a core AppSec enabler, making consistent authorization and policy enforcement easy to adopt without bypassing governance or change control.

DANSKE BANK | Senior Software Engineer

Oct 2020 – May 2021

Brabrand · API Integration Tribe

Legacy modernization, API security, and data integration

At Danske Bank, I worked on modernization initiatives covering both legacy data integration and core banking APIs. The work included a CDC-based proof of concept for migrating data from DB2 on-premises to Kafka, and design of APIs and domain models for new Spring Boot services on OpenShift.

My role: I led technical proof-of-concept work, conducted testing, and helped manage vendor evaluation for the data-integration initiative. In parallel, I contributed to the design of Spring Boot services and architectural blueprints that helped teams migrate from mainframe applications to OpenShift-based services, with risk assessment, security review, and consistent AuthN/AuthZ built into the patterns.

Impact: The work produced reusable foundations for later modernization efforts and clarified viable migration patterns from DB2 and mainframe into Kafka and OpenShift-based services.

ALEXANDRA INSTITUTTET | Senior Security Architect

Oct 2019 – Sep 2020

Aarhus · Security Lab

Security architecture, threat modeling, and IoT/OT standards

Security reviews, threat modeling, and DevSecOps enablement: I worked with Danish companies and public-sector stakeholders on security architecture for connected products and backend services, including threat modeling, architecture and code reviews, and technical security assessments. I also helped teams move from DevOps to DevSecOps using OWASP SAMM, ASVS, SAST, and SCA.

IoT, OT, and standards work: As a subject-matter expert in the Danish Standards working group on cyber and information security, I helped shape Denmark's input to ISO 27000, IEC 62443, and ETSI EN 303 645, with focus on IoT and operational technology. With Aalborg University, I also helped build honeypots targeting IoT and OT environments to study real-world attack vectors.

Digital identity and consent services: As Security Architect and Lead Developer, I designed and implemented a Java/Spring Boot MVP for a privacy-first consent management service integrating with NemID. I also conducted a white-box security review of a Danish digital identity solution, covering the mobile app (including its Kotlin code), backend, authentication flows, and data protection.

DANSKE BANK | Senior Software Engineer

Aug 2018 – Sep 2019

Aarhus

New EDI platform and secure customer onboarding

Modernizing the EDI platform: I contributed to a new Electronic Data Interchange (EDI) platform built in C# on OpenShift, using RabbitMQ for messaging and designed to run alongside the legacy system during a carefully managed transition.

Secure onboarding flow for EDI customers: As lead developer alongside two external consultants, I designed and implemented a secure customer onboarding flow for the EDI platform, migrating from a legacy proprietary encryption format to OpenPGP-based file encryption with secure key management.

TRACKUNIT | Software Engineer

Nov 2017 – Jul 2018

Aalborg

Real-time telemetry platform and migration from legacy applications

I designed and implemented APIs and scalable microservices on AWS for real-time telemetry from connected construction equipment, using Apache Flink, Kafka, and Cassandra to process high volumes of streaming data. Alongside the new platform, I maintained the existing legacy applications and migrated their services onto it, working in both Java and C#.

CRYPTOMATHIC | Software Engineer

Jan 2016 – Oct 2017

Aarhus

Middleware for hardware security modules and qualified e-signatures

I developed and maintained middleware for hardware security modules (HSMs), building web services and APIs for qualified electronic signatures under the eIDAS regulation, with a focus on integration testing and secure, correct cryptographic operations across platforms.

Education

AALBORG UNIVERSITY | PhD in Computer Science

Aug 2012 – Jan 2016

Aalborg

Research focused on the modeling and verification of distributed systems using formal methods. My dissertation, “Synchronization and Control of Quantitative Systems”, was supervised by Prof. Kim G. Larsen and Prof. Jiri Srba and developed mathematical foundations for controlling and synchronizing quantitative systems.

AALBORG UNIVERSITY | MSc in Computer Science and Mathematics

2007 – 2012

Aalborg

Professional activities

EXTERNAL EXAMINER

2018 – Present

Censorkorpset i Datalogi (2018 – Present)
Civilingeniørcensorkorpset (2022 – 2026)

Courses and theses in secure software, static analysis, and distributed systems.

CONFERENCES | KubeCon and Devoxx

KubeCon 2022 – 2026 · Devoxx 2022 – 2024